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Technical overview of nuclear microsatellites for *Fagus* sp., and their utility in *F. sylvatica* from the central Balkans (Serbia)

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ABSTRACT

Selectively neutral nuclear microsatellites (SSRs) are available for *Fagus* sp., and have been used in *F. sylvatica*, an ecologically and economically most important European *Fagus* species. We provide a comprehensive technical overview of available *Fagus* sp. SSRs (185 loci), carry out meta-analysis of loci used in *F. sylvatica* (62 loci employed in 62 surveys), and validate a set of 16 loci in 45 individuals of this species from the central Balkans (Serbia). Erroneous usage of marker's names/authors is rather frequent, and loci successfully used in a number of studies are characterized by other authors by high prevalence of null alleles and even multilocus amplification products. Frequent occurrence of null alleles at FS4-46, used in 26 surveys to date, most likely indicates a failure to record multiple alleles at this locus. Twelve loci are reliable/informative in *F. sylvatica* from the Balkans (5–18 alleles/locus, H_E ranging from 0.523 to 0.850), while four loci are characterized by high prevalence of null alleles (sfc0161 and sfc1063) and multilocus amplification products (FS4-46 and Fagsyl_007038). Our findings are important for future population genetics and studies on adaptation of *F. sylvatica* to its environment, because the latter rely on both selectively neutral and adaptive markers under selection.

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Introduction

Adaptation of forest trees to their usually heterogeneous environments became an attractive and fruitful field of research especially over the past few decades partly because of the need to better understand the response of these long-lived and sessile organisms, which are important from both ecological and economical aspects, to the rapid anthropogenic climate change (IPCC 2014). It has been postulated that possible climate change outcomes of forest trees populations are migration, adaptation and extinction (Aitken et al. 2008). Although the cases of extirpation of forest tree populations/species in response to the global warming have not been recorded to date, examples of climate change-driven migrations are available (e.g. Lescop-Sinclair and Payette 1995; Villalba and Veblen 1998; Sturm et al. 2001; Lloyd and Fastie 2003; Parmesan and Yohe 2003; Millar et al. 2004; Jump and Peñuelas 2005; Walther et al. 2005; Beckage et al. 2008; Bell et al. 2014; Zhang et al. 2018), indicating an onset of the foreseen severe alternation in distributions of all tree species during the next century (Malcolm et al. 2002). However, the pace of the climate change is exceeding by far the migration ability of tree species (Thomas et al. 2004), and thus, standing variation appears to be essential for their local adaptation and survival (Christmas et al. 2016). Available methods for identification of loci related to adaptation, however, are sensitive to evolutionary forces

that can mimic selection, such as population structure (Excoffier et al. 2009; Nielsen et al. 2009; Hoban et al. 2016). Therefore, the usage of molecular markers which are considered selectively neutral (hereafter selectively neutral markers) is still required because they may provide insights into among- and within-population genetic structure, and thus, may be used for the distinction of effects of these phenomena from the effects of selection (e.g. Scalfi et al. 2014; Müller et al. 2015, 2017; Pluess et al. 2016; Müller, Seifert, Lübke, et al. 2017).

Nuclear microsatellites (simple sequence repeats – SSRs), which have been historically considered as selectively neutral, are markers of choice for genotyping in plant and animal species because they are hypervariable, multiallelic, co-dominant, highly reproducible and amenable to automation and high throughput genotyping (Tautz and Schlötterer 1994; Jarne and Lagoda 1996; Schlötterer 1998; Chambers and MacAvoy 2000; Li et al. 2002; Dieringer and Schlötterer 2003; Ellegren 2004; Varshney et al. 2005; Buschiazzi and Gemmell 2006; Oliveira et al. 2006; Selkoe and Toonen 2006; Subirana and Messegueur 2008; Guichoux et al. 2011; Putman and Carbone 2014; Ellegren and Galtier 2016). In comparison to genomic SSRs (gSSRs), which are usually found within non-coding and anonymous regions of the nuclear genome (Morgante et al. 2002; Ellegren 2004) and commonly identified by the screening of cloned libraries

by Sanger sequencing (Schoebel et al. 2013), genic or expressed-sequence-tagged SSRs (EST-SSRs) are found within transcribed but untranslated portion of the genome. EST-SSRs, developed from transcriptome sequences, are characterized by fewer null alleles and lower polymorphism in comparison to gSSRs, but are highly transferable to related species due to the higher level of sequence conservation of transcribed DNA across species (Varshney et al. 2005; Selkoe and Toonen 2006; Zalapa et al. 2012).

The genus *Fagus* L. (beech, Fagaceae) comprises approximately ten monoecious broadleaved tree species distributed in the northern hemisphere (Denk 2003). One of the most studied species of this genus is *Fagus sylvatica* L. which, according to the Flora Europaea (Tutin et al. 1993) and recent systematic revisions (Denk 1999a, 1999b), is a species complex comprising two subspecies, *F. sylvatica* subsp. *sylvatica*, and *F. sylvatica* subsp. *orinetalis* (for information on other taxa within this species complex see Gömöry et al. 2018). The former taxon (denoted hereafter as *F. sylvatica* or common or European beech) is characterized by a wide European distribution, and the latter (denoted hereafter as *F. orientalis* or oriental beech) is found in the southeast Balkan Peninsula, Bulgaria, northern Greece, Caucasus and Alborz mountains in Iran.

Natural populations of *F. sylvatica* have been studied to date by means of nuclear microsatellites by a number of authors, and variability of these markers has been used for assessing, for instance, levels of genetic diversity and genetic structuring among populations (e.g. Vornam 2004; Vornam et al. 2004; Kramer et al. 2010; Rajendra et al. 2014; Sjölund and Jump 2015; Bontemps et al. 2016), within-population fine-scale genetic structure (e.g. Chybicki et al. 2009; Dounavi et al. 2010; Piotti et al. 2013; Rajendra et al. 2014; Sjölund and Jump 2015), demographic history of a species (e.g. Gömöry et al. 2018), pollen flow (e.g. Piotti et al. 2012), mating system (Gauzere et al. 2013) and construction of the genetic linkage map and QTL analysis (Scalfi et al. 2004). Although single nucleotide polymorphisms (SNPs) tend to be used increasingly in studies focusing on adaptation and have been developed for *F. sylvatica* as well (Lesur et al. 2015), on per-locus basis, microsatellites are advantageous over SNPs because of their high allelic diversity and relative ease of transfer between closely related species (Guichoux et al. 2011). Thus, they have also been used in surveys on adaptation of *F. sylvatica* to its environment along with various panels of adaptive loci (e.g. Kramer et al. 2008; Seifert et al. 2012; Csilléry et al. 2014; Müller et al. 2015, 2017; Dounavi et al. 2016; Müller, Seifert, Lübke, et al. 2017). However, different authors employed different microsatellite loci with various success in above mentioned studies, and this highlights the need to provide an overview of nuclear microsatellites that may successfully be used in this species.

The aim of our study is to provide a comprehensive technical overview of nuclear microsatellites currently available for *Fagus* sp. and those used in *F. sylvatica* to date, and to validate the performance of a subset of presumably more informative genomic microsatellite loci in *F. sylvatica* individuals originating from the central Balkans (Serbia). To the authors' best knowledge, beech populations from the Balkans have

not been studied to date at a large scale by means of nuclear microsatellites. Loci selected for validation comprise, on the one hand, those that are most commonly used to date, and, on the other hand, the most recently developed ones.

Materials and methods

Nuclear microsatellites in *Fagus* sp.

All studies in English reporting nuclear SSRs specifically developed for *Fagus* sp. available in PubMed database and Google Scholar on 25 March 2018 were downloaded. Combinations of search words used for the literature search were microsatellite or SSR plus *Fagus* or beech, without constraints on year of publication. The studies were used for assembling a list of all SSRs available at present for *Fagus* sp., and for extracting the data on: (i) a particular *Fagus* species used for marker development; (ii) the number of individuals used for validation of loci; (iii) microsatellite repeat motif; (iv) length or length range of PCR amplification product(s), (v) occurrence of null alleles; (vi) number of alleles (A), and, if available, (vii) gene diversity (H_E).

Usage of nuclear microsatellites in *F. sylvatica*

All studies, Proceeding papers and Meeting reports in English, German, French and Italian citing articles that report the development of nuclear microsatellites in *Fagus* sp. were downloaded, and those in which SSRs were used for various studies in *F. sylvatica* were recorded. Since the authors who employed *Fagus* sp. microsatellites in *F. sylvatica* applied occasionally loci developed for related taxa, these loci were also recorded.

Selection of microsatellites for validation in *F. sylvatica*

In order to validate the performance of published *Fagus* sp. SSRs in *F. sylvatica*, a subset of 16 loci was selected for PCR amplification in this species. They comprised loci which were most commonly used to date and also those which were published most recently and thus, were used in a few studies only. The other criteria used for the selection of loci were: (i) microsatellite structure (perfect SSRs only); (ii) repeat motif (loci with di- and tri-nucleotide motifs only); (iii) occurrence and the frequency of null alleles; (iv) the number of alleles ($A \geq 4$), and, if available, (v) estimated gene diversity ($H_E \geq 0.500$), reported initially by the authors of the loci. However, data on null alleles and H_E were lacking in some publications.

Plant material, DNA extraction, PCR amplification and fragment sizing

The plant material used for microsatellites performance validation comprised 45 individuals of *F. sylvatica* originating from three populations from the central Balkans (Serbia), namely Tara (abbreviation TR), Žagubica (ZG) and Rudnik

(RU). Buds from 13 to 16 individuals per population were collected during the autumn 2016. Upon removal of outer bud scales, buds were dried in silica gel, and kept in freezer prior to the analyses. Up to 30 mg of dried buds from each individual was homogenized with TissueLyser II (Qiagen, Valencia, CA, USA) and used for extraction of the total genomic DNA with DNeasy Plant Mini Kit (Qiagen, Valencia, CA, USA). Genomic DNA was quantified and assessed for purity utilizing NanoVue (GE Healthcare Europe, Freiburg, Germany). DNA solutions were diluted to working concentrations of 50 ng/μl.

Forward (F) primers from each primer pair of all selected loci were labelled with one out of the four different fluorescent dyes from the Dye set G5 (DS-33, Applied Biosystems, USA) to enable fragment sizing on an automated DNA fragment analyser. PCRs were carried out using Multigene Opti Max (Labnet International, Inc.) and following PCR conditions and profiles given originally by the authors of the loci. PCR amplification products were separated commercially by MacroGen Europe, Amsterdam, the Netherlands (<http://dna.macrogen.com/eng/>) via capillary electrophoresis using 96-capillary 3730xl DNA Analyzer automated sequencer (Applied Biosystems, Inc. USA). The lengths of PCR amplification products were assessed with GeneMapper (Applied Biosystems Inc., Foster City, USA).

Validation of the performance of selected microsatellites

The performance of selected SSRs amplified in *F. sylvatica* individuals from the central Balkans (Serbia) was validated based on the success of PCR amplification (the lack of null alleles and multilocus amplification products) and parameters of genetic diversity (A and H_E) estimated using Arlequin 3.5.1.2 (Excoffier et al. 2005).

Results

Nuclear microsatellites in *Fagus* sp.

Until the 25 March 2018, seven publications reporting the development of 78 novel gSSRs for *Fagus* sp. were available in PubMed and Google Scholar (Supplementary Table 1). In addition, primers for amplification of ten novel microsatellite loci in *F. sylvatica* developed by G.G. Vendramin have not been published to date, but have been used by a number of authors. On the other hand, Pastorelli et al. (2003) and Vornam et al. (2004) re-designed/optimized primers of Tanaka et al. (1999) for amplification of mfc5 and mfc11, respectively, for usage in *F. sylvatica* and re-named them to FCM5 and mfs11, respectively. Koch et al. (2009) re-designed primers for amplification of three microsatellite loci of Tanaka et al. (1999), mfc5, mfc7 and mfc11, for usage in *F. grandifolia*, while Hanaoka et al. (2007) improved five primer pairs of Asuka et al. (2004) developed for *F. crenata* for usage in the same species. Therefore, the total number of gSSRs developed for various *Fagus* species was 98. Along with a single study reporting 87 novel EST-SSRs in *F. crenata* (Ueno et al. 2009), the total number of nuclear SSRs loci specifically developed for *Fagus* sp. was 185.

F. sylvatica was the most frequently used species for the development of primers for PCR amplification of microsatellite loci, and primers for amplification of 44 loci are currently available for this species (Pastorelli et al. 2003; Vornam et al. 2004; Lefèvre et al. 2012; Pluess and Määtänen 2013; G.G. Vendramin, pers. comm.). It is followed by *F. crenata* for which primer pairs for amplification of 30 loci have been specifically developed to date (Tanaka et al. 1999; Asuka et al. 2004; Hanaoka et al. 2007), *F. longipetiolata* (11 loci, Liu et al. 2009), *F. hayatae* (10 loci, Ju et al. 2012), *F. japonica* (nine loci, Tanaka et al. 1999), *F. orientalis* (seven loci, Pastorelli et al. 2003), and *F. grandifolia* (three loci, Koch et al. 2009).

Repeat motifs were available for 95 out of 98 reported gSSRs, of which 82 were perfect microsatellites, one interrupted, nine compound and, at three loci, different repeat motifs were in addition interrupted (e.g. interrupted/compound). Majority of perfect gSSRs comprised dinucleotide repeat motifs (69) and trinucleotide repeat motifs (11), while tetranucleotide and hexanucleotide repeat motifs were present in one locus each. The most common dinucleotide motifs in perfect gSSRs were AG (19 loci) and GA (15), followed by TC (12) and CT (10). Motifs CA, TG, AC, AT and GT were present in five, three, two, two and one locus, respectively.

Six loci were characterized by null alleles by their developers, FS1-25 and FS4-46 (Pastorelli et al. 2003) and Fagsyl_001217, Fagsyl_001563, Fagsyl_004597 and Fagsyl_006608 (Pluess and Määtänen 2013).

The number of alleles (A) at a particular locus and gene diversity (H_E) were assessed by the authors of loci based on amplification in at least 20 individuals. A and H_E varied to a great extent among studies: A ranged from two (Fl68, sample size, $N=21$, Liu et al. 2009) to 40 (sfc0161, $N=34$, Asuka et al. 2004), and H_E ranged from 0.216 (mfc4, $N=30$, Tanaka et al. 1999) to 0.980 (sfc0161, $N=34$, Asuka et al. 2004) (Supplementary Table 1). The length of the shortest allele observed at a particular locus was 77 bp (sfc0161, Asuka et al. 2004), and the length of the longest allele was 461 bp (TB-100, Ju et al. 2012).

The only one study available to date reporting the development of EST-SSRs in *Fagus* sp. was that of Ueno et al. (2009). The authors developed 87 loci for *F. crenata* and demonstrated that upon testing these loci in 12–16 individuals of this species, the number of alleles per locus ranged from 2 to 21, and H_E ranged from 0.06 to 0.97. However, only 39% of these loci were perfect microsatellites while the others were interrupted loci, compound loci and the combination of the latter two.

Usage of nuclear microsatellites in *F. sylvatica*

Out of the 98 gSSRs developed/re-designed for various *Fagus* species, 47 loci were used in 62 studies focusing on *F. sylvatica* published over the past two decades (Figure 1, Supplementary Table 1). However, we have to mention that it is possible that certain Proceedings papers and Meeting reports have not been included into our meta-analysis because we were not able to acquire them (e.g. Meeting reports mentioned by the Kramer et al. 2010; Liesebach 2012).

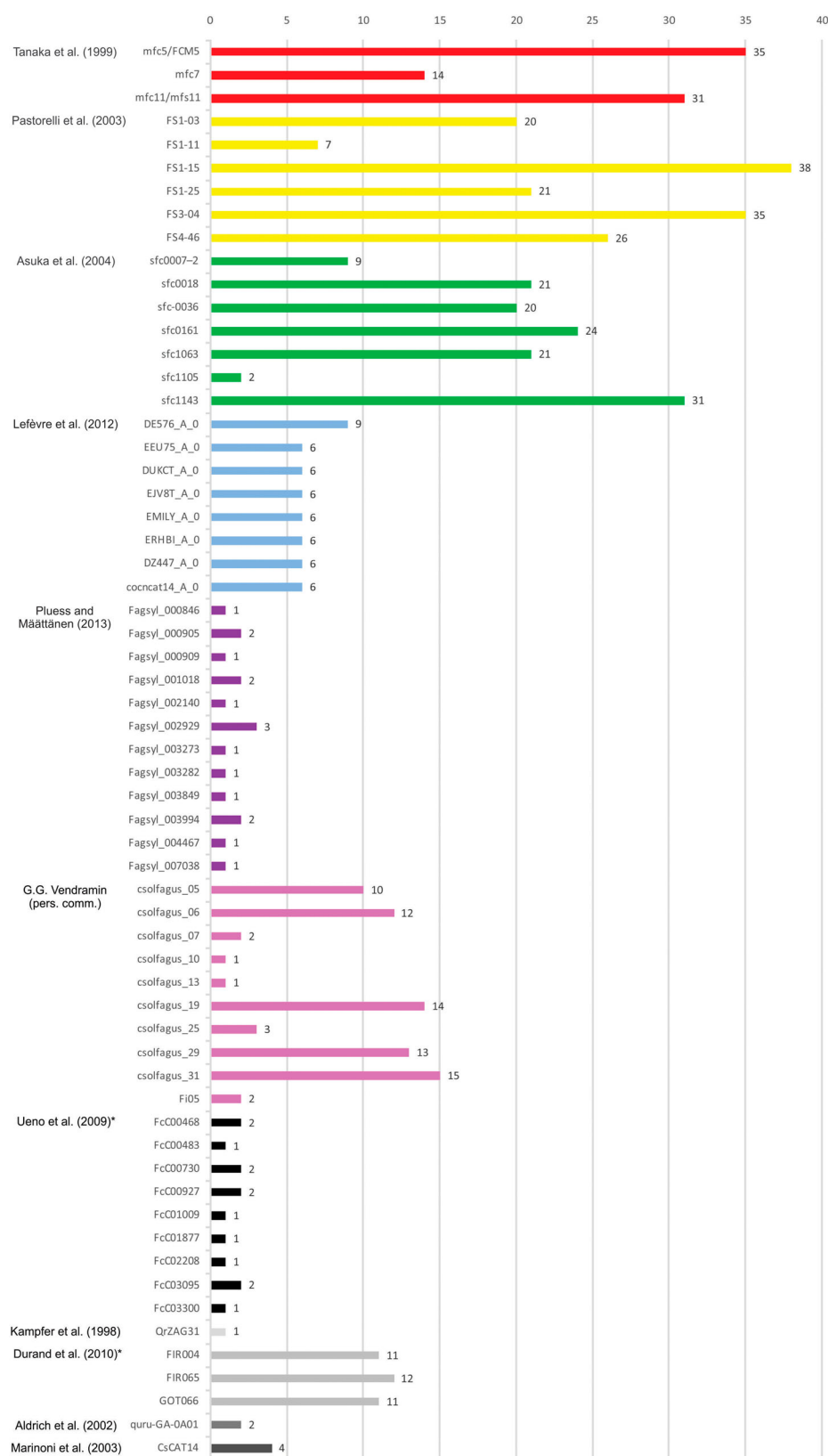


Figure 1. Number of studies in which nuclear microsatellites (genomic and EST-SSRs) developed for *Fagus* sp. and related taxa have been used in *F. sylvatica*. EST-SSRs are labelled with*.

The most commonly used loci were those of Pastorelli et al. (2003), labelled with the prefix FS, which were used in seven (locus FS1-11) to 38 surveys (loci FS1-15). Nine out of the 16 loci developed by Asuka et al. (2004) (prefix sfc), have not

been used to date in *F. sylvatica*, while the remaining loci were employed in up to 31 surveys (sfc1143). All nine loci developed of Tanaka et al. (1999) (prefix mfc) have been tested in at least one study focusing on *F. sylvatica*,

however, only three have been used successfully in a number of studies (but see below). Microsatellites developed for *F. sylvatica* by G.G. Vendramin (unpublished) (nine loci with the prefix csolfagus and a locus Fi05), Lefèvre et al. (2012) (eight loci with no uniform prefixes) and Pluess and Määtänen (2013) (18 loci with the prefix Fagsyl) have been used in up to 15, nine and three studies, respectively. However, along with eight novel loci developed by Lefèvre et al. (2012), these authors reported also the usage of eight loci developed previously by various authors (see Supplementary Table 1). Eleven SSRs developed for *F. longipetiolata* by Liu et al. (2009) and ten loci developed for *F. hayatae* by Ju et al. (2012) have not been employed in *F. sylvatica* to date. It is worth mentioning that Ju et al. (2012) reported cross-species amplification of their loci not only in *F. sylvatica* but also in *F. crenata*, *F. grandifolia*, *F. japonica*, *F. longipetiolata*, *F. lucida* and *F. orientalis*.

Only nine out of the 87 EST-SSRs developed for *F. crenata* by Ueno et al. (2009) (prefix FcC) have been used in *F. sylvatica* to date. Sandurska et al. (2017) employed four EST-SSR loci along with gSSRs, while Dounavi et al. (2016) employed exclusively EST-SSRs in their study (four loci used also by Sandurska et al. 2017, and five additional loci). The majority of EST-SSR loci used by both Sandurska et al. (2017) and Dounavi et al. (2016) were perfect microsatellites with di- and tri-nucleotide repeat motifs.

SSRs developed for related species, *Castanea sativa* Mill., *Quercus robur* L. and *Q. rubra* L. have occasionally been used in *F. sylvatica* as well. They comprise gSSRs CsCAT14 (developed for *C. sativa* locus by Marinoni et al. 2003), QrZAG31 (*Quercus robur*, Kampfer et al. 1998) and quru-GA-0A01 (*Q. rubra* locus, Aldrich et al. 2002) which were used in four, one and two studies, respectively, and EST-SSRs developed for *Q. robur* by Durand et al. (2010) which have been employed in 11 (FIR004 and GOT066) and 12 studies (FIR065) focusing on *F. sylvatica* (Supplementary Table 1).

Therefore, the total number of microsatellite loci successfully used to date in *F. sylvatica* is 62 (47 novel/re-designed gSSRs developed for *Fagus* sp., nine EST-SSRs developed for *F. crenata*, and six loci developed for related species) which have been employed in 62 studies. In addition, six loci of Tanaka et al. (1999) and one locus of Pluess and Määtänen (2013) have been tested in up to four studies, and have been excluded due to the high prevalence of null alleles and/or multilocus amplification products.

The names and the authors of loci used in *F. sylvatica* over the past two decades have been erroneously cited in 29 out of the 62 studies (47%), and that has led to the substantial confusion regarding the usage of loci in a number of surveys. The most problematic was locus mfc5 and, more generally, the remaining mfc loci of Tanaka et al. (1999). Kramer et al. (2008) stated: “we developed new microsatellites (... MFC5 ..., Pastorelli et al. 2003)” during the DynaBeech project, but Pastorelli et al. (2003) named this locus as FCM5 after optimizing primers/procedure for the amplification of the locus mfc5 of Tanaka et al. (1999) for usage in *F. sylvatica*. However, the authors of seven studies list in their material and methods locus FCM5 but state that the authors of this locus are Tanaka et al. (1999). In two studies, Pastorelli et al.

(2003) were cited as the authors of the locus mfc5. Further errors observed in one study each comprise usage of the name FCM5 in the table and MFC5 throughout the text, and the statement that the authors of FCM5 and MFC5 are Pastorelli et al. (2003) and Vornam et al. (2004), respectively. Although Vornam et al. (2004) re-designed primers for the amplification of a single locus of Tanaka et al. (1999), mfc11, for usage in *F. sylvatica*, and re-named it to mfs11, Vornam et al. (2004) have been cited as the authors of various loci such as mcf11, mfc7 or mfs7 (note the misspelling of the prefix comprising three letters). Pastorelli et al. (2003) were cited as the authors of mfc5 or MFC7, and Tanaka et al. (1999) as the authors of mfs11. Erroneous usage of names of loci refer also to the locus FS4-46 (named erroneously as FS1-46 in one study) and sfc0161 (sfc1061, in two studies), while erroneous citing of authors of loci encompass csolfagus loci developed by G.G. Vendramin and co-workers for which Pastorelli et al. (2003) (one study) or Lefèvre et al. (2012) (five studies) were cited as their developers.

On the other hand, the same locus was used twice in some studies because it was amplified in *F. sylvatica* with both original and re-designed primers that gave different estimates of parameters of genetic diversity and genetic differentiation (e.g. mfc5 and FCM5 in the study of Nowakowska and Sulkowska 2015, and mfc11 and mfs11 used in the study of Hasenkamp et al. 2011).

Whenever possible, the presence and frequency of null alleles at SSRs in studies focusing on *F. sylvatica* were recorded. Loci having the frequency of null alleles ≥ 0.07 were delineated as those characterized by the presence of null alleles (Supplementary Table 1). In cases when the frequency of null alleles was provided for each out of the several studied populations, we recorded the presence of null alleles at a locus even if they were observed in a single population. Null alleles were not observed at nine EST-SSRs used to date in *F. sylvatica*, but they were present at 33 out of the 47 gSSRs (78.72%), and at three out of the six loci developed initially for related species and used in common beech (Supplementary Table 1). However, these data have to be taken with cautions because certain loci were used in one study only and they were not characterized by null alleles (e.g. eight Fagsyl loci, Pluess and Määtänen 2013; and one csolfagus locus, G.G. Vendramin, unpublished), while some other loci were used in a number of studies, and null alleles were observed only in a single survey (e.g. sfc1143 used in 31 studies, null alleles reported in only one study). Thus, it is possible that null alleles might be observed at less frequently employed loci in future studies, as well as that records of null alleles at loci which have not been characterized by null alleles in a number of studies may simply indicate certain technical flaws.

Nonetheless, certain loci were characterized by null alleles in more than 55% of studies (e.g. in ten out of 21 studies that employed FS1-25, and in 16 out of 26 studies that used FS4-46). In several studies, high prevalence of null alleles at a particular locus has led to its exclusion from further analyses. The second most common reason for the exclusion of a particular locus from further analyses was the occurrence of multilocus amplification products which, however, have not been reported at the same locus in other studies (e.g. mfc7, FS1-

11). It is worth mentioning that multilocus amplification products were also observed at *mfc7* in another species, *F. crenata* (Vornam 2004).

SSRs loci were usually amplified in *F. sylvatica* individuals via standard (simplex) PCR reactions (one locus-one PCR reaction). Later on, Lefèvre et al. (2012) proposed multiplex PCR reactions for amplification of eight loci simultaneously, i.e. in one PCR reaction, and provided two sets of loci which may be used for such parallel genotyping. These multiplexes were used subsequently by several authors (e.g. Cordier et al. 2012; de Lafontaine et al. 2013; Gömöry et al. 2015; Westergren et al. 2015). On the other hand, authors rather frequently assembled their own multiplexes with loci which they found informative in preliminary testing in their plant material (e.g. Lander et al. 2011; Carsjens et al. 2014; Sjölund and Jump 2015; Bontemps et al. 2016).

Selection of microsatellites for validation in *F. sylvatica*

Given the presumably higher variability of gSSRs over EST-SSRs, and the fact that available EST-SSRs were specifically developed for *F. crenata* and were mainly interrupted, compound or a combination of the two, we selected 16 perfect gSSRs for validation in *F. sylvatica* from the central Balkans (Serbia). They comprised loci which were most commonly used to date in *F. sylvatica*, such as *mfc5* (Tanaka et al. 1999), FS1-03, FS1-15 and FS4-46 (Pastorelli et al. 2003), *sfc0018*, *sfc0161*, *sfc1063* and *sfc1143* (Asuka et al. 2004), *DES576_A_0* (Lefèvre et al. 2012), *csolfagus_06*, *csolfagus_19* and *csolfagus_31* (G.G. Vendramin, pers. comm.), as well as those which were published more recently and have been used in a few surveys only, such as *Fagsyl_000905*,

Fagsyl_003273, *Fagsyl_003994* and *Fagsyl_007038* (Pluess and Määtänen 2013) (Table 1). Three loci (FS4-46, *DES576_A_0* and *Fagsyl_000905*) comprised trinucleotide repeat motifs while the remaining thirteen loci comprised dinucleotide repeat motifs. The most common dinucleotide repeat motif was AG or GA (nine loci), while motifs CT, TC, GT and TG were present in one locus each.

According to the data provided in all studies that employed these loci in *F. sylvatica*, null alleles were rather common at loci FS4-46 and *mfc5*/FCM5 (reported in 54% and 46% of studies, respectively). Null alleles were also observed at loci FS1-03 (30% of studies), FS1-15 (24%), *DES576_A_0* (22%), *sfc1063* (19%) and *sfc0018* (9%). Locus *Fagsyl_000905* was employed in two studies only, and, in one of them, it was characterized by the presence of null alleles. Null alleles were reported at loci *sfc0161*, *sfc1143*, *csolfagus_06*, *csolfagus_19* and *csolfagus_31* in a single study each (these loci were used in 12–31 surveys). Loci *Fagsyl_003273*, *Fagsyl_003994*, *Fagsyl_007038*, employed in one or two studies, were null-allele free.

Validation of the performance of selected microsatellites

Out of the 16 selected gSSRs, two loci, *sfc0161* and *sfc1063*, failed to generate PCR amplification products in majority of 45 *F. sylvatica* individuals originating from three populations from the central Balkans (Serbia) while more than two alleles per locus per individual were observed in all individuals at locus *Fagsyl_007038* and in 25 individuals at locus FS4-46. Therefore, the former two loci were characterized as loci with null alleles, and the latter two as loci that generate multilocus amplification products in studied plant material. A specific

Table 1. Genetic diversity parameters (*A*, number of alleles per locus, *H_E*, expected heterozygosity) at 12 genomic microsatellites successfully used in 45 *Fagus sylvatica* individuals from three populations (ZG, TR and RU) from the central Balkans (Serbia).

#	Locus	<i>F. sylvatica</i> from the central Balkans (Serbia)					References
		Pop <i>N</i>	ZG 13	TR 16	RU 16	TOTAL 45	
1	<i>csolfagus_06</i>	<i>A</i>	8	7	7	9	G.G. Vendramin (pers. comm.)
		<i>H_E</i>	0.784	0.844	0.803	0.810	
2	<i>csolfagus_19</i>	<i>A</i>	9	8	8	11	G.G. Vendramin (pers. comm.)
		<i>H_E</i>	0.805	0.678	0.768	0.750	
3	<i>csolfagus_31</i>	<i>A</i>	8	9	9	11	G.G. Vendramin (pers. comm.)
		<i>H_E</i>	0.784	0.854	0.818	0.819	
4	<i>DES576_A_0*</i>	<i>A</i>	4	6	5	6	Lefèvre et al. (2012)
		<i>H_E</i>	0.725	0.777	0.781	0.761	
5	<i>Fagsyl_000905*</i>	<i>A</i>	5	3	4	5	Pluess and Määtänen (2013)
		<i>H_E</i>	0.675	0.365	0.662	0.567	
6	<i>Fagsyl_003273</i>	<i>A</i>	4	5	4	5	Pluess and Määtänen (2013)
		<i>H_E</i>	0.485	0.477	0.607	0.523	
7	<i>Fagsyl_003994</i>	<i>A</i>	7	7	8	10	Pluess and Määtänen (2013)
		<i>H_E</i>	0.799	0.797	0.785	0.794	
8	FS1-03	<i>A</i>	10	10	9	14	Pastorelli et al. (2003)
		<i>H_E</i>	0.846	0.701	0.678	0.742	
9	FS1-15	<i>A</i>	8	9	8	10	Pastorelli et al. (2003)
		<i>H_E</i>	0.846	0.771	0.838	0.818	
10	<i>sfc0018</i>	<i>A</i>	9	9	7	11	Asuka et al. (2004)
		<i>H_E</i>	0.642	0.820	0.600	0.687	
11	<i>sfc1143</i>	<i>A</i>	8	6	9	10	Asuka et al. (2004)
		<i>H_E</i>	0.831	0.715	0.842	0.796	
12	<i>mfc5</i>	<i>A</i>	12	11	9	18	Tanaka et al. (1999)
		<i>H_E</i>	0.867	0.832	0.852	0.850	

All loci are perfect microsatellites with di-nucleotide and tri-nucleotide repeat motifs (the latter are labelled with*). Pop, population, *N*, number of individuals.

pattern of peak distribution was observed in several electropherograms of the locus FS4-46, i.e. after the first peak of expected length and suitable height (above the threshold height of 500 relative fluorescence units, RFU), numerous peaks of expected length but below the threshold height were recorded (Supplementary Figure 1). The number of alleles in each individual at the remaining 12 loci was one or two, as expected in diploid organisms, and all electropherograms were easily readable.

The number of alleles per locus in overall sample ranged from five (Fagsyl_000905 and Fagsyl_003273) to 18 (mfc5), and H_E from 0.523 (Fagsyl_003273) to 0.850 (mfc5) (Table 1).

Discussion

Increasing pace of the global climate change (IPCC 2014) highlights the need to better understand the adaptation of forest tree species to their environments, and studies of this kind became more abundant especially over the past two decades (e.g. Howe et al. 2003; Aitken et al. 2008; Kramer et al. 2008; Neale and Ingvarsson 2008; Kremer et al. 2012; Seifert et al. 2012; Alberto et al. 2013; Savolainen et al. 2013; Sork et al. 2013; Scalfi et al. 2014; Müller et al. 2015, 2017; Dounavi et al. 2016; Hoban et al. 2016; Müller, Seifert, Lübke, et al. 2017; Housset et al. 2018). Selectively neutral molecular markers are used in studies of this kind as well because, based on available methods, identification of loci related to adaptation require the parallel usage of both adaptive and selectively neutral markers in order to delineate the effects of selection from those caused by evolutionary forces, such as population structure, assessed by markers not affected by various selective pressures (Excoffier et al. 2009; Nielsen et al. 2009; Hoban et al. 2016). Although new sequencing techniques facilitate genome-wide approaches to study genetic variation in numerous species including *F. sylvatica* (e.g. Lesur et al. 2015), selectively neutral molecular markers which are still most commonly used in various studies in plants and animals are nuclear microsatellites (Tautz and Schlötterer 1994; Jarne and Lagoda 1996; Schlötterer 1998; Chambers and MacAvoy 2000; Li et al. 2002; Dieringer and Schlötterer 2003; Ellegren 2004; Varshney et al. 2005; Buschiazzi and Gemmell 2006; Oliveira et al. 2006; Selkoe and Toonen 2006; Subirana and Messeguer 2008; Guichoux et al. 2011; Putman and Carbone 2014; Ellegren and Galtier 2016). We provide a comprehensive technical overview of available nuclear SSRs in *Fagus* sp. and their usage in *F. sylvatica*, and validate a set of loci in individuals of this species originating from the central Balkans (Serbia) which have not been studied to date at a large scale with these markers.

Nuclear microsatellites in *Fagus* sp.

Tanaka et al. (1999) published the first set of nine gSSRs specifically developed for *Fagus* sp. The authors used random amplified hybridization microsatellites method to identify SSR regions in the genome of *F. crenata* along with southern hybridization used to detect CT/GA-repeat loci, and applied developed loci in a closely related species,

F. japonica. During the next two decades, 69 additional novel gSSRs (Pastorelli et al. 2003; Asuka et al. 2004; Liu et al. 2009; Ju et al. 2012; Lefèvre et al. 2012; Pluess and Määtänen 2013) and 87 EST-SSRs (Ueno et al. 2009) have been developed for *Fagus* sp. (Supplementary Table 1). In addition, ten gSSRs developed for *F. sylvatica* have been used in a number of studies but have not been published to date (G.G. Vendramin, pers. comm.), while ten loci developed for one *Fagus* species were re-designed for the same or alternative species by a number of authors (e.g. Pastorelli et al. 2003; Vornam et al. 2004; Hanaoka et al. 2007; Koch et al. 2009). Therefore, the number of nuclear SSRs specifically developed for *Fagus* sp. is 185 in total. Given the number of microsatellites developed for other tree species, such as *Eucalyptus* sp. (more than 700; see Grattapaglia et al. 2015) and, in particular, in agronomically important crops (> 27.000 microsatellite loci in the *Vitis vinifera* L. genome, Di Gaspero et al. 2007; Cipriani et al. 2008; > 4.000 polymorphic loci for *Triticaceae* sp., Deng et al. 2016; > 350 loci for *Vicia faba* L., reviewed by Duc et al. 2015), this number is relatively modest.

Although *F. sylvatica* was not the first species for which nuclear gSSRs were developed, the highest number of loci (44) is currently available for this species (Pastorelli et al. 2003; Lefèvre et al. 2012; Pluess and Määtänen 2013; G.G. Vendramin, pers. comm.). The number of loci specifically developed for other *Fagus* species ranges from three (*F. grandifolia*, Koch et al. 2009) to 30 (*F. crenata*, Tanaka et al. 1999; Asuka et al. 2004; Hanaoka et al. 2007). The highest number of *Fagus* sp. gSSRs are perfect microsatellites (84%) of which majority (84%) comprise dinucleotide repeat motifs. Although the most common dinucleotide motifs in these loci were AG (19 loci), GA (15), TC (12) and CT (10), the prevalence of these motifs does not correspond to the abundance of various microsatellite repeat motifs in plant genomes in which AA/TT and AT/TA repeat motifs prevail (Lagercrantz et al. 1993). This is the results of the methodology employed for the development of loci in *Fagus* sp., because the authors mainly targeted CT/GA-repeat SSRs (e.g. Tanaka et al. 1999; Pastorelli et al. 2003).

Important parameters that usually guide workers during the selection of nuclear SSRs for testing and further usage in their plant or animal material are the number of alleles at a particular locus (A) and, if available, gene diversity (H_E). According to the A and H_E reported by the authors of 98 gSSRs developed for *Fagus* sp., some loci may indeed be characterized as highly informative even upon amplification in 20–30 individuals, for instance, locus sfc0161 (Asuka et al. 2004) that harbours 40 alleles and displays H_E of 0.980. (Supplementary Table 1). However, A and H_E are, among other factors, dependent on the sample size and demographic histories of populations. Thus, it is not surprising that loci initially characterized by relatively low A turn out to be rather informative in a larger sample of the same species and *vice versa*, and the same holds for H_E . On the other hand, EST-SSRs developed for *F. crenata* by Ueno et al. (2009) and tested in 16 individuals were expectedly somewhat less polymorphic in comparison to gSSRs (Varshney et al. 2005; Selkoe and Toonen 2006; Zalapa et al. 2012). However, it has been shown that certain EST-SSRs may be more variable than

gSSRs (Aleksić et al. 2009), and this indeed holds for several *F. crenata* EST-SSRs, such as FcC01877 (16 alleles, $H_E = 0.96$) and FcC01463 (12 alleles, $H_E = 0.91$).

Usage of nuclear microsatellites in *F. sylvatica*

Over the past two decades, 47 out of the 98 gSSRs developed/re-designed for *Fagus* species (48%), nine out of the 87 EST-SSRs developed for *F. crenata* (10%) and six loci developed for related species were successfully used in 62 studies focusing on *F. sylvatica* (Supplementary Table 1). The most commonly employed loci in general were those of Pastorelli et al. (2003) used in seven (locus FS1-11) to 38 surveys (FS1-15) (Figure 1, Supplementary Table 1). Certain loci developed by other authors have also been used in a high number of studies, for instance, mfc5 (Tanaka et al. 1999)/FCM5 (Pastorelli et al. 2003) which was used in 35 surveys, and sfc1143 of Asuka et al. (2004) used in 31 surveys. It is important to mention that erroneous usage of marker's names/authors has been observed in a number of studies, and that the most problematic was locus mfc5 (Tanaka et al. 1999) for which Pastorelli et al. (2003) re-designed and optimized primers for usage for *F. sylvatica*, and re-named it to FCM5. The most frequently used loci developed by G.G. Vendramin and co-workers (unpublished) and Lefèvre et al. (2012) were csolfagus_31 (15 studies) and DE576_A_0 (nine surveys), respectively. Loci developed for *F. sylvatica* by Pluess and Määtänen (2013) have been used to date in up to three surveys, while those of Liu et al. (2009) and Ju et al. (2012), developed for *F. longipetiolata* and *F. hayatae*, respectively, have not been employed in *F. sylvatica*.

EST-SSRs developed for *F. crenata* by Ueno et al. (2009), on the other hand, have only sporadically been employed in *F. sylvatica* (Figure 1, Supplementary Table 1). This may be due to the fact that 61% of these loci were interrupted, compound, or loci that have different repeat motifs separated by a various number of nucleotides. At compound loci, the distinction between variability in the repeat copy number of two motifs harbouring the same number of nucleotides may be achieved only by sequencing. Nonetheless, according to the A and H_E values reported in *F. sylvatica* at EST-SSR loci used by Sandurska et al. (2017) and Dounavi et al. (2016), of which majority are perfect microsatellites with di- and tri-nucleotide repeat motifs, it appears that certain EST-SSRs developed for *F. crenata* may indeed be suitable and informative for further usage in *F. sylvatica*. Expectedly, EST-SSR loci employed in *F. sylvatica* were characterized by less null alleles compared to gSSRs (Varshney et al. 2005; Selkoe and Toonen 2006; Zalapa et al. 2012).

Genomic SSRs developed for *Castanea sativa* (Marinoni et al. 2003) and *Quercus robur* (Kampfer et al. 1998; Durand et al. 2010) as well as EST-SSRs developed for *Q. rubra* (Aldrich et al. 2002) have also been successfully used in several studies in *F. sylvatica*. However, A and H_E at these loci were usually lower than at loci specifically developed for *Fagus* sp. or *F. sylvatica* (e.g. Scalfi et al. 2005; Rajendra et al. 2014; Müller and Finkeldey 2016). In addition, some of these loci were characterized by null alleles (e.g. FIR004, Rajendra et al. 2014, and FIR065, Müller 2013). This is in

accordance with the view of Chybicki and Burczyk (2009) that non-native loci may sometimes be scored incorrectly and thus influence the estimates of null allele frequencies. Utility of 20 novel EST-SSRs in *F. sylvatica*, which were developed for *Quercus rubra* and successfully cross-amplified in common beech (Müller and Gailing 2018), is yet to be assessed.

Null alleles, defined as alleles that constantly fail to amplify to detectable levels via PCR (see Dakin and Avise 2004 for a review), are recessive and undetected in heterozygotes, and thus, may affect estimation of allele frequencies (Nascimento de Sousa et al. 2005; Chybicki and Burczyk 2009), inbreeding coefficient (Chybicki and Burczyk 2009), F_{ST} and genetic distances (Chapuis and Estoup 2007), parentage analysis (Dow and Ashley 1998; Dakin and Avise 2004; Oddou-Muratorio et al. 2009) and Bayesian assignment testing (Falush et al. 2007; Carlsson 2008). Possible reasons for the occurrence of null alleles at a microsatellite locus are poor primer annealing due to nucleotide sequence divergence, differential amplification of size-variant alleles, PCR failure due to inconsistent DNA template quality or low template quality (Dakin and Avise 2004). However, we suggest that at least some loci characterized by high prevalence of null alleles may actually represent loci that generate multilocus amplification products and at which multiple alleles were simply undetected previously. This was the case with FS4-46 which was characterized by high frequency of null alleles in 16 studies (Supplementary Table 1) while we observed multiple alleles at this locus (Supplementary Figure 1). This locus as well as FS1-25 were characterized by null alleles by their developers (Pastorelli et al. 2003), and they displayed null alleles in 55% of surveys in which they were employed (Supplementary Table 1). Four Fagsyl loci characterized also by null alleles by the developers (Pluess and Määtänen 2013), however, have not been used in *F. sylvatica* except for Fagsyl_004597 which has been tested and subsequently excluded from further analyses in one study only (Cvrčková et al. 2017). The remaining gSSRs used in *F. sylvatica* to date have sporadically been characterized by null alleles in local plant material in some but not in all studies in which they were employed (e.g. FS1-03, sfc0018, sfc1063).

Validation of the performance of the selected microsatellites

The validation of 16 gSSRs in *F. sylvatica* from the central Balkans revealed that: (i) two loci, sfc0161 and sfc1063 (Asuka et al. 2004), which have successfully been used in more than 20 studies in this species to date, are characterized by high prevalence of null alleles in our local plant material, and should not be used in future large scale studies with plant material from this region; (ii) two loci, of which one (FS4-46, Pastorelli et al. 2003) was used in 26 surveys and was characterized by null alleles in more than a half of these studies, and the other (Fagsyl_007038, Pluess and Määtänen 2013), which was used successfully in one study (Pluess et al. 2016), are characterized by multilocus amplification products; (iii) high prevalence of null alleles at a particular locus in a number of studies may simply indicate failure to detect

multilocus amplification products at that locus, such as, for instance FS4-46. This questions results of previous studies in which this locus was successfully used; (iv) 12 gSSRs, of which some are amongst the most commonly used loci in *F. sylvatica* and the others have been used in this species less frequently, are suitable for further studies in this species from the central Balkans (A ranging from five to 18, H_E ranging from 0.523 to 0.850 upon amplification in 45 individuals originating from three populations). Given the various range of allele lengths at these 12 reliable and informative gSSRs delineated in this work, they would indeed be amenable to parallel automated genotyping via multiplexing, which is a labour and cost-effective procedure compared to the standard simplex genotyping (e.g. Lander et al. 2011; Lefèvre et al. 2012; Carsjens et al. 2014; Sjölund and Jump 2015; Bontemps et al. 2016).

Conclusions

We demonstrate that the number of nuclear SSRs developed for *Fagus* sp. is relatively modest (185 loci), and that 62 loci developed for *Fagus* sp. and related species has been employed to date in 62 various studies in *F. sylvatica*, which is ecologically and economically the most important *Fagus* species in Europe. We observed erroneous usage of marker's names/authors in a number of studies, of which the most common cases refer to mfc5 (Tanaka et al. 1999)/FCM5 (Pastorelli et al. 2003) and other mfc loci of Tanaka et al. (1999), as well as high prevalence of null alleles at a number of loci of which some were used in up to 26 studies. Multilocus amplification products have been reported less frequently at rather low number of loci, and we suggest that this may be due, at least in some cases, to inability to detect multiple alleles at a particular locus. Although gSSRs have predominantly been used in *F. sylvatica* to date, several EST-SSRs developed for *F. crenata* appear to be reliable and informative in this species as well. Out of the 16 gSSRs validated in 45 individuals of *F. sylvatica* from the Balkans, four are not suitable for further usage in this plant material due to the high prevalence of null alleles and occurrence of multilocus amplification products. Our work represents an important contribution with respect to the availability of molecular tools for utilization of selectively neutral nuclear diversity in *F. sylvatica*. This is essential also for future studies on adaptation of this species to its environment especially in terms of the global climate change (e.g. Kramer et al. 2008; Seifert et al. 2012; Csilléry et al. 2014; Scalfi et al. 2014; Müller et al. 2015, 2017; Dounavi et al. 2016; Müller, Seifert, Lübke, et al. 2017).

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Disclosure statement

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References

- Aitken SN, Yeaman S, Holliday JA, Wang T, Curtis-McLane S. 2008. Adaptation, migration or extirpation: climate change outcomes for tree populations. *Evol Appl.* 1:95–111.
- Alberto FJ, Aitken SN, Alía R, González-Martínez SC, Hänninen H, Kremer A, Lefèvre F, Lenormand T, Yeaman S, Whetten R, et al. 2013. Potential for evolutionary responses to climate change—evidence from tree populations. *Glob Chang Biol.* 19:1645–1661.
- Aldrich PR, Michler CH, Sun W, Romero-Severson J. 2002. Microsatellite markers for northern red oak (Fagaceae: *Quercus rubra*). *Mol Ecol Notes.* 2:472–474.
- Aleksić JM, Schueler S, Mengl M, Geburek T. 2009. EST-SSRs developed for other *Picea* species amplify in *Picea omorika* and reveal high genetic variation in two natural populations. *Belg J Bot.* 142:89–95.
- Asuka Y, Tani N, Tsumura Y, Tomaru N. 2004. Development and characterization of microsatellite markers for *Fagus crenata* Blume. *Mol Ecol Notes.* 4:101–103.
- Beckage B, Osborne B, Gavin DG, Pucko C, Siccama T, Perkins T. 2008. A rapid upward shift of a forest ecotone during 40 years of warming in the Green Mountains of Vermont. *Proc Natl Acad Sci.* 105:4197–4202.
- Bell DM, Bradford JB, Lauenroth WK. 2014. Early indicators of change: divergent climate envelopes between tree life stages imply range shifts in the western United States. *Global Ecol Biogeogr.* 23:168–180.
- Bontemps A, Lefèvre F, Davi H, Oddou-Muratorio S. 2016. In situ marker-based assessment of leaf trait evolutionary potential in a marginal European beech population. *J Evol Biol.* 29:514–527.
- Buschiazzo E, Gemmell NJ. 2006. The rise, fall and renaissance of microsatellites in eukaryotic genomes. *Bioessays.* 28:1040–1050.
- Carlsson J. 2008. Effects of microsatellite null alleles on assignment testing. *J Hered.* 99:616–623.
- Carsjens C, Nguyen Ngoc Q, Guzy J, Knutzen F, Meier IC, Müller M, Finkeldey R, Leuschner C, Polle A. 2014. Intra-specific variations in expression of stress-related genes in beech progenies are stronger than drought-induced responses. *Tree Physiol.* 34:1348–1361.
- Chambers GK, MacAvoy ES. 2000. Microsatellites: consensus and controversy. *Comp Biochem Physiol B Biochem Mol Biol.* 126:455–476.
- Chapuis MP, Estoup A. 2007. Microsatellite null alleles and estimation of population differentiation. *Mol Biol Evol.* 24:621–631.
- Christmas MJ, Breed MF, Lowe AJ. 2016. Constraints to and conservation implications for climate change adaptation in plants. *Conserv Genet.* 17:305–320.
- Chybicki IJ, Burczyk J. 2009. Simultaneous estimation of null alleles and inbreeding coefficients. *J Hered.* 100:106–113.
- Chybicki IJ, Trojankiewicz M, Oleksa A, Działuk A, Burczyk J. 2009. Isolation-by-distance within naturally established populations of European beech (*Fagus sylvatica*). *Botany.* 87:791–798.
- Cipriani G, Marrazzo MT, Di Gasparo G, Pfeiffer A, Morgante M, Testolin R. 2008. A set of microsatellite markers with long core repeat optimized for grape (*Vitis* spp.) genotyping. *BMC Plant Biol.* 8:127.
- Cordier T, Robin C, Capdevielle X, Desprez-Loustau ML, Vacher C. 2012. Spatial variability of phyllosphere fungal assemblages: genetic distance predominates over geographic distance in a European beech stand (*Fagus sylvatica*). *Fungal Ecol.* 5:509–520.
- Csilléry K, Lalagüe H, Vendramin GG, González-Martínez SC, Fady B, Oddou-Muratorio S. 2014. Detecting short spatial scale local

- adaptation and epistatic selection in climate-related candidate genes in European beech (*Fagus sylvatica*) populations. *Mol Ecol*. 23:4696–4708.
- Cvrčková H, Máchová P, Poláková L, Trčková O. 2017. Evaluation of the genetic diversity of selected *Fagus sylvatica* L. populations in the Czech Republic using nuclear microsatellites. *J For Sci*. 63:53–61.
- Dakin EE, Avise JC. 2004. Microsatellite null alleles in parentage analysis. *Heredity*. 93:504–509.
- de Lafontaine G, Ducousso A, Lefèvre S, Magnanou E, Petit RJ. 2013. Stronger spatial genetic structure in recolonized areas than in refugia in the European beech. *Mol Ecol*. 22:4397–4412.
- Deng P, Wang M, Feng K, Cui L, Tong W, Song W, Nie X. 2016. Genome-wide characterization of microsatellites in Triticeae species: abundance, distribution and evolution. *Sci Rep*. 6:32224.
- Denk T. 1999a. The taxonomy of *Fagus* in western Eurasia, 1: *Fagus sylvatica* subsp. *orientalis* (= *F. orientalis*). *Feddes Repert*. 110:177–200.
- Denk T. 1999b. The taxonomy of *Fagus* in western Eurasia, 2: *Fagus sylvatica* subsp. *sylvatica*. *Feddes Repert*. 110:381–412.
- Denk T. 2003. Phylogeny of *Fagus* L. (Fagaceae) based on morphological data. *Plant Syst Evol*. 240:55–81.
- Dieringer D, Schlötterer C. 2003. Two distinct modes of microsatellite mutation processes: evidence from the complete genomic sequences of nine species. *Genome Res*. 13:2242–2251.
- Di Gaspero G, Cipriani G, Adam-Blondon AF, Testolin R. 2007. Linkage maps of grapevine displaying the chromosomal locations of 420 microsatellite markers and 82 markers for R-gene candidates. *Theor Appl Genet*. 114:1249–1263.
- Dounavi A, Koutsias N, Ziehe M, Hattemer HH. 2010. Spatial patterns and genetic structures within beech populations (*Fagus sylvatica* L.) of forked and non-forked individuals. *Eur J For Res*. 129:1191–1202.
- Dounavi A, Netzer F, Celepirovic N, Ivanković M, Burger J, Figueroa AG, Schön S, Simon J, Cremer E, Fussi B, et al. 2016. Genetic and physiological differences of European beech provenances (*F. sylvatica* L.) exposed to drought stress. *For Ecol Manage*. 361:226–236.
- Dow BD, Ashley MV. 1998. High levels of gene flow in bur oak revealed by paternity analysis using microsatellites. *J Hered*. 89:62–70.
- Duc G, Aleksić JM, Marget P, Mikić A, Paull J, Redden RJ, Sass O, Stoddard FL, Vandenberg A, Vishniakova M, Torres AM. 2015. Faba bean. In: De Ron AM, editor. Grain legumes, handbook of plant breeding series. New York: Springer + Business Media; p. 141–179.
- Durand J, Bodénès C, Chancerel E, Frigerio JM, Vendramin GG, Sebastiani F, Buonamici A, Gailing O, Koelewijn HP, Villani F, et al. 2010. A fast and cost-effective approach to develop and map EST-SSR markers: oak as a case study. *BMC Genomics*. 11:570.
- Ellegren H. 2004. Microsatellites: simple sequences with complex evolution. *Nature Rev Genet*. 5:435–445.
- Ellegren H, Galtier N. 2016. Determinants of genetic diversity. *Nat Rev Genet*. 17:422–433.
- Excoffier L, Hofer T, Foll M. 2009. Detecting loci under selection in a hierarchically structured population. *Heredity*. 103:285–298.
- Excoffier L, Laval G, Schneider S. 2005. Arlequin ver. 3.0: an integrated software package for population genetics data analysis. *Evol Bioinform*. 1:47–50.
- Falush D, Stephens M, Pritchard JK. 2007. Inference of population structure using multilocus genotype data: dominant markers and null alleles. *Mol Ecol Notes*. 7:574–578.
- Gauzere J, Klein EK, Oddou-Muratorio S. 2013. Ecological determinants of mating system within and between three *Fagus sylvatica* populations along an elevational gradient. *Mol Ecol*. 22:5001–5015.
- Gömöry D, Ditmarová L, Hrivnák M, Jamnická G, Kmeť J, Krajmerová D, Kurjak D. 2015. Differentiation in phenological and physiological traits in European beech (*Fagus sylvatica* L.). *Eur J For Res*. 134:1075–1085.
- Gömöry D, Paule L, Macejovsky V. 2018. Phylogeny of beech in western Eurasia as inferred by approximate Bayesian computation. *Acta Soc Bot Pol*. 87:3582.
- Grattapaglia D, Mamani EM, Silva-Junior OB, Faria DA. 2015. A novel genome-wide microsatellite resource for species of *Eucalyptus* with linkage-to-physical correspondence on the reference genome sequence. *Mol Ecol Resour*. 15:437–448.
- Guichoux E, Lagache L, Wagner S, Chaumeil P, Léger P, Lepais O, Lepoittevin C, Malausa T, Revardel E, Salin F, Petit RJ. 2011. Current trends in microsatellite genotyping. *Mol Ecol Resour*. 11:591–611.
- Hanaoka S, Yuzurihara J, Asuka Y, Tomaru N, Tsumura Y, Kakubari Y, Mukai Y. 2007. Pollen-mediated gene flow in a small, fragmented natural population of *Fagus crenata*. *Botany*. 85:404–413.
- Hasenkamp N, Ziegenhagen B, Mengel C, Schulze L, Schmitt HP, Liepelt S. 2011. Towards a DNA marker assisted seed source identification: a pilot study in European beech (*Fagus sylvatica* L.). *Eur J For Res*. 130:513–519.
- Hoban S, Kelley JL, Lotterhos KE, Antolin MF, Bradburd G, Lowry DB, Poss ML, Reed LK, Storfer A, Whitlock MC. 2016. Finding the genomic basis of local adaptation: pitfalls, practical solutions, and future directions. *The Am Nat*. 188:379–397.
- Housset JM, Nadeau S, Isabel N, Depardieu C, Duchesne I, Lenz P, Girardin MP. 2018. Tree rings provide a new class of phenotypes for genetic associations that foster insights into adaptation of conifers to climate change. *New Phytol*. 218:630–645.
- Howe GT, Aitken SN, Neale DB, Jermstad KD, Wheeler NC, Chen TH. 2003. From genotype to phenotype: unraveling the complexities of cold adaptation in forest trees. *Can J Bot*. 81:1247–1266.
- IPCC. 2014. Summary for policymakers. In: Field CB, Barros VR, Dokken DJ, Mach KJ, Mastrandrea MD, Bilir TE, Chatterjee M, Ebi KL, Estrada YO, Genova RC, et al., editors. Climate change 2014: impacts, adaptation, and vulnerability. Part A: global and sectoral aspects. Contribution of working group I to the fifth assessment report of the intergovernmental panel on climate change. New York: Cambridge University Press; p. 1–32.
- Jarne P, Lagoda P. 1996. Microsatellites, from molecules to populations and back. *Trends Ecol Evol*. 11:424–429.
- Ju LP, Shih HC, Chiang YC. 2012. Microsatellite primers for the endangered beech tree, *Fagus hayatae* (Fagaceae). *Am J Bot*. 99:e453–e456.
- Jump AS, Peñuelas J. 2005. Running to stand still: adaptation and the response of plants to rapid climate change. *Ecol Lett*. 8:1010–1020.
- Kampfer S, Lexer C, Glössl J, Steinkellner H. 1998. Characterization of (GA)n microsatellite loci from *Quercus robur*. *Heredity*. 129:183–186.
- Koch JL, Carey DW, Mason ME. 2009. Use of microsatellite markers in an American beech (*Fagus grandifolia*) population and paternity testing. *Silvae Genet*. 59:62–68.
- Kramer K, Buiteveld J, Forstreuter M, Geburek T, Leonardi S, Menozzi P, Povillon F, Schelhaas MJ, Teissier du Cros E, Vendramin GG, et al. 2008. Bridging the gap between ecophysiological and genetic knowledge to assess the adaptive potential of European beech. *Ecol Modell*. 216:333–353.
- Kramer K, Degen B, Buschbom J, Hickler T, Thuiller W, Sykes MT, de Winter W. 2010. Modelling exploration of the future of European beech (*Fagus sylvatica* L.) under climate change—range, abundance, genetic diversity and adaptive response. *For Ecol Manage*. 259:2213–2222.
- Kremer A, Ronce O, Robledo-Arnuncio JJ, Guillaume F, Bohrer G, Nathan R, Bridle JR, Gomulkiewicz R, Klein EK, Ritland K, et al. 2012. Long-distance gene flow and adaptation of forest trees to rapid climate change. *Ecol Lett*. 15:378–392.
- Lagercrantz U, Ellegren H, Andersson L. 1993. The abundance of various polymorphic microsatellite motifs differs between plants and vertebrates. *Nucl Acids Res*. 21:1111–1115.
- Lander TA, Oddou-Muratorio S, Prouillet-Leplat H, Klein EK. 2011. Reconstruction of a beech population bottleneck using archival demographic information and Bayesian analysis of genetic data. *Mol Ecol*. 20:5182–5196.
- Lefèvre S, Wagner S, Petit RJ, De Lafontaine G. 2012. Multiplexed microsatellite markers for genetic studies of beech. *Mol Ecol Resour*. 12:484–491.
- Lescop-Sinclair K, Payette S. 1995. Recent advance of the arctic treeline along the eastern coast of Hudson Bay. *J Ecol*. 83:929–936.
- Lesur I, Bechade A, Lalanne C, Klopp C, Noirot C, Leplé JC, Le Provost G. 2015. A unigene set for European beech (*Fagus sylvatica* L.) and its use to decipher the molecular mechanisms involved in dormancy regulation. *Mol Ecol Resour*. 15:1192–1204.

- Li YC, Korol AB, Fahima T, Beiles A, Nevo E. 2002. Microsatellites: genomic distribution, putative functions and mutational mechanisms: a review. *Mol Ecol*. 11:2453–2465.
- Liesebach H. 2012. Genotypisierung mit nuklearen Mikrosatellitenmarkern – Möglichkeiten der Datenauswertung am Beispiel von Buchenpopulationen (*Fagus sylvatica* L.) aus einem Herkunftsversuch. *Landbauforsch*. 62:221–236.
- Liu M, Shi MM, Liu MH, Chen XY. 2009. Isolation and characterization of microsatellite loci in *Fagus longipetiolata* Seem. (Fagaceae). *Conserv Genet*. 10:1981–1983.
- Lloyd AH, Fastie CL. 2003. Recent changes in treeline forest distribution and structure in interior Alaska. *Ecoscience*. 10:176–185.
- Malcolm JR, Markham A, Neilson RP, Garaci M. 2002. Estimated migration rates under scenarios of global climate change. *J Biogeogr*. 29:835–849.
- Marinoni D, Akkac A, Bounous G, Edwards KJ, Botta R. 2003. Development and characterization of microsatellite markers in *Castanea sativa* (Mill.). *Mol Breed*. 11:127–136.
- Millar CL, Westfall RD, Delany DL, King JC, Graumlich LJ. 2004. Response of subalpine conifers in the Sierra Nevada, California, USA, to 20th-century warming and decadal climate variability. *Arct Antarct Alp Res*. 36:181–200.
- Morgante M, Hanafey M, Powell W. 2002. Microsatellites are preferentially associated with nonrepetitive DNA in plant genomes. *Nature Genet*. 30:194–200.
- Müller M. 2013. A candidate gene-based association study to investigate potentially adaptive genetic variation in European beech (*Fagus sylvatica* L.) [dissertation]. Germany: Georg-August-Universität Göttingen.
- Müller M, Finkeldey R. 2016. Genetic and adaptive trait variation in seedlings of European beech provenances from Northern Germany. *Silvae Genet*. 65:65–73.
- Müller M, Gailing O. 2018. Characterization of 20 new EST-SSR markers for northern red oak (*Quercus rubra* L.) and their transferability to *Fagus sylvatica* L. and six oak species of section *Lobatae* and *Quercus*. *Ann For Res*. 61:211–222.
- Müller M, Seifert S, Finkeldey R. 2015. A candidate gene-based association study reveals SNPs significantly associated with bud burst in European beech (*Fagus sylvatica* L.). *Tree Genet Genom*. 11:116.
- Müller M, Seifert S, Finkeldey R. 2017. Comparison and confirmation of SNP-bud burst associations in European beech populations in Germany. *Tree Genet Genom*. 13:59.
- Müller M, Seifert S, Lübke T, Leuschner C, Finkeldey R. 2017. De novo transcriptome assembly and analysis of differential gene expression in response to drought in European beech. *PLoS ONE*. 12:e0184167.
- Nascimento de Sousa S, Finkeldey R, Gailing O. 2005. Experimental verification of microsatellite null alleles in Norway spruce (*Picea abies* [L.] Karst.): implications for population genetic studies. *Plant Mol Biol Rep*. 23:113–119.
- Neale DB, Ingvarsson PK. 2008. Population, quantitative and comparative genomics of adaptation in forest trees. *Curr Opin Plant Biol*. 11:149–155.
- Nielsen R, Hubisz MJ, Hellmann I, Torgerson D, Andres AM, Albrechtsen A, Gutenkunst R, Adams MD, Cargill M, Boyko A, et al. 2009. Darwinian and demographic forces affecting human protein coding genes. *Genome Res*. 19:838–849.
- Nowakowska JA, Sulkowska MK. 2015. Capillary electrophoresis as useful tool in analysis of *Fagus sylvatica* L. population genetic dynamics. In: Field effect electroosmosis - a novel phenomenon in electrokinetics and its applications in capillary electrophoresis. London: InTech; p. 49–68.
- Oddou-Muratorio S, Vendramin GG, Buiteveld J, Fady B. 2009. Population estimators or progeny tests: what is the best method to assess null allele frequencies at SSR loci? *Conserv Genet*. 10:1343–1347.
- Oliveira EJ, Pádua JG, Zucchi MI, Vencovsky R, Vieira MLC. 2006. Origin, evolution and genome distribution of microsatellites. *Genet Mol Biol*. 29:294–307.
- Parmesan C, Yohe G. 2003. A globally coherent fingerprint of climate change impacts across natural systems. *Nature*. 421:37–42.
- Pastorelli R, Smulders MJM, Van't Westende WPC, Vosman B, Giannini R, Vettori C, Vendramin GG. 2003. Characterization of microsatellite markers in *Fagus sylvatica* L. and *Fagus orientalis* Lipsky. *Mol Ecol Notes*. 3:76–78.
- Piotti A, Leonardi S, Buiteveld J, Geburek T, Gerber S, Kramer K, Vettori C, Vendramin GG. 2012. Comparison of pollen gene flow among four European beech (*Fagus sylvatica* L.) populations characterized by different management regimes. *Heredity*. 108:322–331.
- Piotti A, Leonardi S, Heuertz M, Buiteveld J, Geburek T, Gerber S, Kramer K, Vettori C, Vendramin GG, Ingvarsson PK. 2013. Within-population genetic structure in beech (*Fagus sylvatica* L.) stands characterized by different disturbance histories: does forest management simplify population substructure? *PLoS ONE*. 8:e73391.
- Pluess AR, Frank A, Heiri C, Lagüë H, Vendramin GG, Oddou-Muratorio S. 2016. Genome-environment association study suggests local adaptation to climate at the regional scale in *Fagus sylvatica*. *New Phytol*. 210:589–601.
- Pluess AR, Määttänen K. 2013. Characterization of eighteen novel microsatellite markers and multiplex PCR protocol for *Fagus sylvatica*. *Conserv Genet Resour*. 5:311–314.
- Putman AJ, Carbone I. 2014. Challenges in analysis and interpretation of microsatellite data for population genetic studies. *Ecol Evol*. 4:4399–4428.
- Rajendra KC, Seifert S, Prinz K, Gailing O, Finkeldey R. 2014. Subtle human impacts on neutral genetic diversity and spatial patterns of genetic variation in European beech (*Fagus sylvatica*). *For Ecol Manage*. 319:138–149.
- Sandurska E, Ulaszewski B, Burczyk J. 2017. Genetic insights into ecological succession from oak (*Quercus robur* L.) to beech (*Fagus sylvatica* L.) dominated forest stands. *Acta Biol Crac Ser Bot*. 59:23–33.
- Savolainen O, Lascoux M, Merilä J. 2013. Ecological genomics of local adaptation. *Nature Rev Genet*. 14:807–820.
- Scalfi M, Leonardi S, Piovani P, Piotti A, Menozzi P. 2005. Fragmentation effects on beech genetic structure. *Ital J For Mount Environm*. 60:352–365.
- Scalfi M, Mosca E, Di Pierro EA, Troggio M, Vendramin GG, Sperisen C, La Porta N, Neale DB, Papa R. 2014. Micro- and macro-geographic scale effect on the molecular imprint of selection and adaptation in Norway spruce. *PLoS ONE*. 9:e115499.
- Scalfi M, Troggio M, Piovani P, Leonardi S, Magnaschi G, Vendramin GG, Menozzi P. 2004. A RAPD, AFLP and SSR linkage map, and QTL analysis in European beech (*Fagus sylvatica* L.). *Theor Appl Genet*. 108:433–441.
- Schlötterer C. 1998. Genome evolution: are microsatellites really simple sequences? *Curr Biol*. 8:132–134.
- Schoebel CN, Brodbeck S, Buehler D, Cornejo C, Gajurel J, Hartikainen H, Keller D, Leys M, Řičanová Š, Segelbacher G, et al. 2013. Lessons learned from microsatellite development for nonmodel organisms using 454 pyrosequencing. *J Evol Biol*. 26:600–611.
- Seifert S, Vornam B, Finkeldey R. 2012. DNA sequence variation and development of SNP markers in beech (*Fagus sylvatica* L.). *Eur J For Res*. 131:1761–1770.
- Selkoe KA, Toonen RJ. 2006. Microsatellites for ecologists: a practical guide to using and evaluating microsatellite markers. *Ecol Lett*. 9:615–629.
- Sjölund MJ, Jump AS. 2015. Coppice management of forests impacts spatial genetic structure but not genetic diversity in European beech (*Fagus sylvatica* L.). *For Ecol Manage*. 336:65–71.
- Sork VL, Aitken SN, Dyer RJ, Eckert AJ, Legendre P, Neale DB. 2013. Putting the landscape into the genomics of trees: approaches for understanding local adaptation and population responses to changing climate. *Tree Genet Genomes*. 9:901–911.
- Sturm M, Racine C, Tape K. 2001. Climate change: increasing shrub abundance in the Arctic. *Nature*. 411:546–547.
- Subirana JA, Messeguer X. 2008. Structural families of genomic microsatellites. *Gene*. 408:124–132.
- Tanaka K, Tsumura Y, Nakamura T. 1999. Development and polymorphism of microsatellite markers for *Fagus crenata* and the closely related species, *F. japonica*. *Theor Appl Genet*. 99:11–15. doi:10.1007/s001220051203.
- Tautz D, Schlötterer C. 1994. Simple sequences. *Curr Opin Genet Dev*. 4:832–837.

- Thomas CD, Cameron A, Green RE, Bakkenes M, Beaumont LJ, Collingham YC, Erasmus BFN, de Siqueira MF, Grainger A, Hannah L, et al. 2004. Extinction risk from climate change. *Nature*. 427:145–148.
- Tutin TG, Burges NA, Chater AO, Edmondson JR, Heywood VH, Moore DM, Valentine DH, Walters SM, Webb DA. 1993. *Flora Europaea*. 2nd ed. Vol. 1. Cambridge: Cambridge University Press.
- Ueno S, Taguchi Y, Tomaru N, Tsumura Y. 2009. Development of EST-SSR markers from an inner bark cDNA library of *Fagus crenata* (Fagaceae). *Conserv Genet*. 10:1477–1485.
- Varshney RK, Graner A, Sorrells ME. 2005. Genic microsatellite markers in plants: features and applications. *Trends Biotechnol*. 23:48–55.
- Villalba R, Veblen TT. 1998. Influences of large-scale climatic variability on episodic tree mortality in northern Patagonia. *Ecology*. 79:2624–2640.
- Vornam B. 2004. Identifizierung von Buchenherkünften (*Fagus sylvatica* L.) mittels DNA-Markern. Herkunftssicherung und Zertifizierung von forstlichem Vermehrungsgut. 54:93–98.
- Vornam B, Decarli N, Gailing O. 2004. Spatial distribution of genetic variation in a natural beech stand (*Fagus sylvatica* L.) based on microsatellite markers. *Conserv Genet*. 5:561–570.
- Walther GR, Berger S, Sykes MT. 2005. An ecological ‘footprint’ of climate change. *Proc Roy Soc B Biol Sci*. 272:1427–1432.
- Westergren M, Božić G, Ferreira A, Kraigher H. 2015. Insignificant effect of management using irregular shelterwood system on the genetic diversity of European beech (*Fagus sylvatica* L.): a case study of managed stand and old growth forest in Slovenia. *For Ecol Manage*. 335:51–59.
- Zalapa JE, Cuevas H, Zhu H, Steffan S, Senalik D, Zeldin E, McCown B, Harbut R, Simon P. 2012. Using next-generation sequencing approaches to isolate simple sequence repeat (SSR) loci in the plant sciences. *Am J Bot*. 99:193–208.
- Zhang T, Niinemets Ü, Sheffield J, Lichstein JW. 2018. Shifts in tree functional composition amplify the response of forest biomass to climate. *Nature*. 556:99–102.